

The RESOLVE Swiss trial for people with chronic low back pain: statistical analysis plan for a cluster randomised controlled trial

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Abstract

Background. Statistical analysis plans describe the planned data management and analysis of a clinical trial before the data are seen, and so protect the eventual results from data-driven analytical choices. This paper reports the statistical analysis plan for the RESOLVE Swiss trial, which evaluates graded sensorimotor retraining combined with pain neuroscience education against usual physiotherapy for people with chronic low back pain.

Design and analysis. RESOLVE Swiss is a two-arm parallel-group *cluster* randomised controlled trial in Swiss physiotherapy practices; the practice is the unit of randomisation and the patient is the unit of analysis. The primary outcome is back-specific disability, measured with the Roland–Morris Disability Questionnaire at 18 weeks after randomisation. The primary analysis is a linear mixed model with a random intercept for practice and adjustment for the baseline value of the outcome, estimating the participant-average difference between arms with Kenward–Roger degrees of freedom. Because the trial randomises a small number of practices, and because mixed models are known to produce anti-conservative inference in that setting, we prespecify two further analyses of the primary outcome: a cluster-level (two-stage) analysis and an exact randomisation test over the finite set of possible allocations. We report the intra-cluster correlation with its uncertainty, the analyses of secondary outcomes, the handling of missing

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data, non-adherence and contamination, and the planned reporting.

Conclusion. Prespecifying the analysis reduces the scope for bias in the results of RESOLVE Swiss and makes explicit how the small number of clusters is dealt with.

Trial registration. [OPEN: registry and registration number]

Keywords: Low back pain; Cluster randomised trial; Statistical analysis plan; Mixed models; Physical therapy.

1 Introduction

1.1 Background and rationale

Chronic low back pain is among the largest contributors to years lived with disability worldwide, and the effects of the treatments physiotherapists currently offer are, on average, modest. The Australian RESOLVE trial tested whether a treatment aimed at the central nervous system representation of the back — graded sensorimotor precision training preceded by pain neuroscience education — reduces pain more than a sham intervention, and reported a small but consistent benefit at 18 weeks [1]. An earlier and much smaller Swiss pilot compared a related programme with usual physiotherapy and pointed in the same direction without being able to resolve the question [2].

Two things are unresolved. First, RESOLVE compared the intervention with a sham; the clinically relevant comparison for Swiss practice is against usual physiotherapy as it is actually delivered. Second, the intervention has so far been delivered in a trial setting by trial therapists. RESOLVE Swiss therefore delivers it in ordinary physiotherapy practices, by the therapists who work there. That choice makes the trial pragmatic, and it forces the design to be a *cluster* randomised trial: once a practice has been trained in the intervention, its therapists cannot un-learn it for the next patient, so allocation has to happen at the level of the practice rather than the patient [3].

Cluster randomisation changes the statistics in ways that are easy to get wrong, and the difficulty is sharpened here because the number of practices is small. This analysis plan is written before the analysis of outcome data and follows the recommendations for the content of statistical analysis plans in clinical trials [4], together with the CONSORT extension for cluster

randomised trials [3]. It reports the planned frequentist analysis of the primary and secondary outcomes. [OPEN: decide whether the Bayesian interim analysis is reported in this paper or in a companion paper; see section 3.8]

2 Methods

2.1 Trial design

RESOLVE Swiss is a two-arm, parallel-group, cluster randomised controlled trial with 1:1 intended allocation at the level of the physiotherapy practice. Practices are the clusters; patients treated within a practice are the units of observation and of analysis. Outcome data are collected at baseline and at 18 weeks after randomisation [OPEN: confirm the full follow-up schedule; the Australian trial also collected 26 and 52 weeks].

At the time of writing, [OPEN: confirm] 5 practices are allocated to the intervention arm and 9 to the control arm. The consequences of that split, and of any change to it, are quantified in section 2.6.

Ethical approval was granted by [OPEN: ethics committee and approval number].

2.2 Eligibility

A cluster randomised trial has two eligibility layers, and both are reported here because selection can operate at either [3].

Practices. [OPEN: practice-level inclusion criteria: e.g. minimum patient volume, number of therapists, willingness to be randomised to either arm, completion of the intervention training]

Participants. [OPEN: participant-level criteria; from the sample size work the trial requires a baseline RMDQ of at least 7 points, please confirm and complete the remaining criteria]

2.3 Interventions

[OPEN: brief description of the intervention and of usual physiotherapy as the comparator, with a reference to the protocol; note the number of sessions and the training the intervention therapists received]

Usual physiotherapy is expected to be heterogeneous between practices. What control practices

actually deliver will be recorded and described, because in a pragmatic trial the comparator is part of the finding rather than a nuisance.

2.4 Randomisation and allocation

[OPEN: describe how the practices were allocated, who generated the sequence, and whether allocation was simple, stratified, or covariate-constrained]

Two points matter for the analysis. First, any variable used to stratify or constrain the allocation must be adjusted for in the analysis, otherwise the standard errors are wrong [5]. Second, if practices were recruited in waves and later waves were allocated non-randomly, wave and treatment arm would be confounded; recruitment wave will therefore be recorded for every practice and, if practices entered in distinct waves, carried in the model as a covariate.

Participants cannot be blinded to the treatment they receive. Outcome questionnaires are self-reported. [OPEN: state who is blinded — data manager, statistician — and confirm that the statistician will be blinded to arm labels until the primary analysis is complete]

2.5 Recruitment of participants after cluster allocation

Because practices are randomised before their patients are recruited, and because the recruiting therapists know which arm their practice is in, patient selection can differ between arms. This identification or recruitment bias is the characteristic threat to cluster randomised trials and is not present in individually randomised trials [3, 6]. We will therefore report, by arm and by practice, the number of patients screened, found eligible, approached, and consenting, and we will compare baseline characteristics between arms with this threat explicitly in mind. Baseline imbalance in a cluster randomised trial with few clusters is expected by chance and is not by itself evidence of bias; it is, however, the main signal that recruitment differed between arms, and it will be interpreted alongside the screening counts rather than by significance testing [3, 7].

2.6 Sample size

The target difference is 2 points on the Roland–Morris Disability Questionnaire (RMDQ, range 0–24), the smallest between-group difference the investigators consider clinically worthwhile in

this population. The between participant standard deviation of the RMDQ at 18 weeks is taken from the Australian trial (intervention 3.6, SD 4.6; control 6.4, SD 5.8 [1]), giving a pooled SD of 5.2 points. With a two-sided $\alpha = 0.05$ and 80% power, an individually randomised trial would need 108 participants per arm; adjusting the 18-week score for its baseline value, with an assumed baseline-to-follow-up correlation of 0.6, reduces this to 69 per arm.

Clustering inflates this requirement by the design effect. Allowing for unequal practice sizes [8],

$$DE = 1 + \left[\left(cv^2 \frac{k-1}{k} + 1 \right) \bar{m} - 1 \right] \rho, \quad (1)$$

where $\bar{m}$ is the mean number of analysed participants per practice, k the number of practices, ρ the intra-cluster correlation and cv the coefficient of variation of practice size. Intra-cluster correlations reported for primary care outcomes are usually small, with a median near 0.01 [9]. Unequal cluster size can be ignored when $cv < 0.23$, but for trials randomising general practices cv is typically around 0.65 [8]; physiotherapy practices are the closest analogue we have, so we plan on $cv = 0.65$ and report the more optimistic 0.4 as well. With $\bar{m} = 14$, $k = 14$ and $cv = 0.65$, the design effect is 1.19 at $\rho = 0.01$, 1.56 at $\rho = 0.03$ and 1.93 at $\rho = 0.05$, so the requirement is 82, 107 and 133 participants per arm respectively. The trial therefore aims for 100 analysed participants per arm, which corresponds to 110 recruited per arm allowing 10% loss to follow-up (Table 1); this covers an intra-cluster correlation up to about 0.03.

Two approximations in this calculation should be named. The variance reduction from baseline adjustment is applied as the individual-level factor $1 - r^2$ and the clustering inflation as (1); a formula that handles baseline adjustment inside a cluster randomised design, using the individual-level and the cluster-level baseline correlations separately, is available and gives a slightly smaller requirement when the cluster-level correlation is high [10]. Our version is therefore mildly conservative. Second, the intra-cluster correlation is assumed, not known; the calculation is reported across a range of values rather than at a single one, which is what sensitivity to this assumption requires [11].

Three features of the design as it currently stands should be stated plainly rather than buried. First, with 14 practices the between-cluster component of the variance rests on 14 numbers, and the relevant reference distribution has roughly 12 degrees of freedom; power is correspondingly lower than the design effect alone suggests. Holding the analysed sample at 200 participants

and using a t reference distribution on $k_1 + k_2 - 2$ degrees of freedom, the power to detect a 2-point difference is 77% at $\rho = 0.01$, 65% at $\rho = 0.03$ and 56% at $\rho = 0.05$.

Second, rebalancing the existing practices from 5 versus 9 to 7 versus 7 recovers only about 3 percentage points of power at every intra-cluster correlation considered, because the total analysed sample is what it is. Equal allocation of clusters is the efficient choice when the intra-cluster correlation and the total variance are the same in both arms, and a deliberate imbalance is justified only when they are not [12]; we have no reason to expect them to differ here, so 7 versus 7 is preferable, but the efficiency argument for changing the split is weak.

Third — and this is the binding constraint — *the number of practices is too small*. Five clusters in one arm sits at the bottom of the range in which the recommended analyses have been evaluated [13], and a minimum of about ten clusters per arm has been recommended for cluster randomised trials generally, with the practical rule of adding two to three clusters per arm to any sample size derived from a normal approximation [14]. Few clusters also raise the probability of chance imbalance on practice characteristics and reduce the number of analyses available [15]. Adding practices while holding practice size at 14 participants raises power from 62% with 5 per arm to 80% with 7 per arm and 90% with 9 per arm at $\rho = 0.01$ (Table 2) — a much better return than adding patients to the practices already enrolled, because increasing cluster size at a fixed number of clusters has strongly diminishing returns [16]. If additional practices are recruited, they must be randomised as a batch rather than assigned to the intervention arm, so that the new wave contains its own randomised contrast [17, 18].

All sample size and power computations are reproducible from the script `R/sample_size.R` that accompanies this paper.

2.7 Data integrity and missing data

[OPEN: describe the data capture system (REDCap), double entry or electronic capture, and the data-cleaning steps]

Data will be inspected for implausible values, range violations and duplicate records before the analysis begins, and the cleaned data set will be locked before any comparison between arms is made. We will report the amount and the pattern of missing outcome data by arm and by practice, since in a cluster trial an entire practice can be lost, which is a different problem from individual dropout.

2.8 Analytic principles

- Participants will be analysed in the arm to which their practice was allocated, regardless of the treatment actually delivered (intention to treat).
- All treatment effects will be reported as point estimates with 95% confidence intervals. Two-sided tests at a nominal α of 0.05 accompany the primary and secondary analyses, but the interpretation will rest on the estimate and its interval rather than on whether a threshold was crossed [19, 20].
- Confidence intervals and p values will not be adjusted for multiplicity. There is one primary outcome at one time point; secondary outcomes are explicitly labelled as secondary and will be interpreted as such [4].
- Every analysis accounts for clustering by practice. No analysis will treat patients from the same practice as independent.
- Analyses will be carried out in R [21], with mixed models fitted using `lme4` [22] and small-sample inference obtained with `lmerTest` and `pbkrtest` [23, 24]. Analysis code will be published with the results.

2.9 Outcome definitions

2.9.1 Primary outcome

Back-specific disability at 18 weeks after randomisation, measured with the Roland–Morris Disability Questionnaire, a 24-item instrument scored from 0 (no disability) to 24 [25, 26].

2.9.2 Secondary outcomes

[OPEN: list secondary outcomes and their instruments and time points, e.g. pain intensity (NRS), quality of life (EQ-5D-5L), global perceived effect, adverse events, credibility/expectancy, and any process measures]

3 Analysis

3.1 Baseline description

We will describe the sample at two levels, as the CONSORT cluster extension requires [3]. At the practice level (Table 3, panel A) we will report the number of practices per arm, the number of therapists, practice size, urban or rural setting, and the number of participants contributed. At the participant level (panel B) we will report means and standard deviations for approximately normally distributed continuous variables, medians and interquartile ranges otherwise, and counts with percentages for categorical variables. Baseline characteristics will not be tested for statistical significance between arms: the null hypothesis of no difference is true by construction under randomisation, so such a test answers no question anyone is asking [7, 27].

3.2 Primary outcome

3.2.1 Estimand

The quantity of interest is the *participant-average* difference in mean RMDQ at 18 weeks between the two arms: the difference that would be seen if every patient of this kind in these practices were treated one way rather than the other, with each patient counting equally. In a cluster randomised trial with unequally sized clusters this is not the same quantity as the cluster-average effect, in which each practice counts equally, and different analyses target different quantities [28]. The choice is made here, at the design stage, as the consensus extension of the ICH E9(R1) estimand addendum to cluster randomised trials requires [29]. We prespecify the participant-average effect and report the cluster-average effect alongside it.

One consequence deserves stating, because it constrains how the primary model below may be read. A mixed model with a random intercept weights clusters by their inverse variance, which is neither participant-average nor cluster-average weighting; when cluster size is informative — that is, when outcomes or treatment effects vary systematically with practice size — such a model targets neither estimand cleanly [28]. The cluster-level analyses specified in section 3.2.3 recover each estimand explicitly through their weighting, which is a second reason to report them rather than treating them as a mere robustness check. We will report the distribution of

214 practice sizes so that readers can judge how much this matters here.

215 3.2.2 Primary model

216 Let y_{ij} denote the 18-week RMDQ score of participant i in practice j , y_{ij}^0 the same participant's
217 baseline score, and $x_j = 1$ if practice j was allocated to the intervention and 0 otherwise. The
218 primary analysis is the linear mixed model

$$219 \quad y_{ij} = \beta_0 + \beta_1 x_j + \beta_2 y_{ij}^0 + u_j + \varepsilon_{ij}, \quad u_j \sim N(0, \tau^2), \quad \varepsilon_{ij} \sim N(0, \sigma^2), \quad (2)$$

220 where u_j is a random intercept for practice and β_1 is the treatment effect. Analysis of
221 covariance on the follow-up score is preferred to analysing change scores: it is unbiased under
222 randomisation, it is more precise whenever baseline and follow-up are positively correlated, and
223 it does not inherit the regression-to-the-mean behaviour of change scores [30, 31]; for cluster
224 randomised trials specifically, analysis of covariance and the equivalent constrained-baseline
225 longitudinal formulation are the recommended ways to use a pre-randomisation measurement
226 of the outcome [32]. The mixed-model analysis of covariance estimator is moreover consistent
227 under misspecification of the working model, and with few clusters the main-effects form used
228 here is preferable to one carrying treatment-by-covariate interactions, which spends degrees of
229 freedom and requires a robust variance estimator [33].

230 Covariate adjustment in a cluster randomised trial is not automatically beneficial, unlike in an
231 individually randomised one: each cluster-level covariate costs a degree of freedom, so that the
232 degrees of freedom available for the treatment effect are approximately the number of practices
233 minus two minus the number of cluster-level covariates [34]. With 14 practices the budget is 12,
234 and we therefore adjust for the individual baseline score — an individual-level covariate, which
235 does not consume cluster-level degrees of freedom — and for nothing else except the variables
236 used to stratify or constrain the randomisation, which must be adjusted for or the type I error
237 rate is not maintained [5, 11].

238 The model will be fitted by restricted maximum likelihood. Confidence intervals and p values
239 for β_1 will use the Kenward–Roger approximation to the degrees of freedom and the associated
240 small-sample adjustment to the covariance matrix of the fixed effects [35], which is the standard
241 remedy for the anti-conservatism of naive mixed-model inference when the number of clusters is

small [13, 36].

3.2.3 Inference with a small number of clusters

Uncorrected mixed models and generalised estimating equations have inflated type I error rates when the number of clusters is small, and the inflation is substantial: in a review and reanalysis of published cluster randomised trials, a large share of trials with few or moderately many clusters had used methods that do not maintain the nominal error rate [15, 36]. In a simulation across designs with 4 to 40 clusters, Leyrat and colleagues found that for continuous outcomes the Satterthwaite and Kenward–Roger corrections to a mixed model, and variance-weighted cluster-level analysis, did maintain the nominal rate, whereas uncorrected approaches did not [13]. Model (2) with Kenward–Roger degrees of freedom is therefore a defensible primary analysis for the RMDQ, and it is what the current guidance for cluster trials with fewer than about 40 clusters recommends [11, 37]. It is nonetheless an asymptotic argument resting on 14 numbers, so we prespecify two further analyses of the primary outcome, reported alongside the model-based result rather than instead of it. Cluster-level analyses, by contrast, need no such correction at all, which is why they are the natural companion when the number of clusters is very small [37].

Cluster-level analysis. We will regress y_{ij} on y_{ij}^0 within practices, take the practice-level mean residual as a single summary per practice, and compare the 14 summaries between arms with a two-sample t test on $k_1 + k_2 - 2 = 12$ degrees of freedom, weighting practices by their analysed size for the participant-average estimand and unweighted for the cluster-average estimand [28, 38]. This two-stage analysis gives up some efficiency but its type I error rate is correct by construction with any number of clusters.

Exact randomisation test. Because randomisation acts on practices, the set of allocations that could have occurred is finite and small: with 5 of 14 practices allocated to the intervention there are exactly $\binom{14}{5} = 2002$ of them, and with a 7 versus 7 split, 3432. We will therefore enumerate the full set, recompute the cluster-level test statistic under each allocation, and take the proportion of allocations giving a statistic at least as extreme as the observed one as an exact p value. This test makes no distributional assumption and is valid however few clusters there are [13, 39]. Its resolution is limited by the design: the smallest attainable two-sided p value with a 5 versus 9 split is $2/2002 \approx 0.001$. If randomisation was covariate-constrained, the

enumeration will be restricted to the allocations that satisfy the same constraints, since only those could have been selected [40].

If the three analyses agree, the model-based estimate is reported as the primary result. If they disagree, the disagreement itself will be reported and the randomisation test treated as the arbiter of the significance claim, since it is the only one of the three whose validity does not depend on the number of clusters.

3.2.4 Intra-cluster correlation

We will report the estimated intra-cluster correlation for the primary outcome, $\hat{\rho} = \hat{\tau}^2 / (\hat{\tau}^2 + \hat{\sigma}^2)$, with a 95% confidence interval obtained by profile likelihood, and, if the interval is very wide, we will say so rather than presenting the point estimate alone. Reporting the observed intra-cluster correlation is a CONSORT requirement for cluster trials and is what makes the next trial in this field easier to design [3, 9].

3.3 Supportive analysis over both time points

To keep the results comparable with the Australian trial [41], we will additionally fit a mixed model for repeated measures using all measurement occasions, with fixed effects for arm, time, and their interaction, a random intercept for practice and a random intercept for participant nested within practice. The treatment effect at 18 weeks is the corresponding arm-by-time contrast. This analysis is supportive; (2) remains the primary one. Mixed models for repeated measures behave acceptably in cluster randomised trials with continuous outcomes missing at random [42].

3.4 Secondary outcomes

Secondary outcomes will be analysed with models of the same structure as (2), with the link function and distribution appropriate to the outcome, always including a random intercept for practice and always using the same small-sample correction. Effects will be reported as point estimates with 95% confidence intervals (Table 5). No formal claim of efficacy will be based on a secondary outcome.

3.5 Sensitivity analyses

The following are prespecified and will be reported whether or not they change the conclusion.

1. **Leave-one-practice-out.** The primary model refitted 14 times, each time omitting one practice. With few clusters a single atypical practice can carry the result, and the spread of the 14 estimates shows how much it does.
2. **Therapist effects.** A third level for therapist within practice. This is the more faithful description of the data hierarchy, but variance components at three levels are poorly identified with 14 practices, so it is a sensitivity analysis rather than the primary model.
3. **Unadjusted analysis.** Model (2) without the baseline covariate, to show what the baseline adjustment contributes.
4. **Cluster-mean baseline.** Model (2) with the practice mean baseline score added as a further fixed effect, which separates the within-practice from the between-practice association with baseline disability.
5. **Wave.** If practices entered in distinct recruitment waves, the primary model with wave as a fixed effect.

3.6 Missing data

The primary analysis uses all participants with a recorded 18-week outcome and is valid under a missing-at-random assumption conditional on the baseline score and the covariates in the model. We will report the number and proportion of missing outcomes by arm and by practice, and compare the baseline characteristics of participants with and without an 18-week outcome. As a sensitivity analysis we will use multiple imputation by chained equations with the practice included in the imputation model, so that the imputation respects the clustering [42]. If an entire practice is lost, that will be reported separately and not imputed.

3.7 Non-adherence and contamination

Adherence will be summarised as the number of intervention sessions attended and as the proportion of participants attending at least [OPEN: threshold] of them. If non-adherence is

substantial, we will estimate the complier-average causal effect using allocation as an instrument. With 14 clusters this analysis is imprecise, and it is prespecified as exploratory. Contamination — control practices adopting elements of the intervention — is a real possibility in a trial where practices know each other. [OPEN: describe how contamination is measured, e.g. a therapist questionnaire at the end of the trial]. Any contamination detected will be described and its likely direction of effect discussed; it biases the estimated effect towards the null [43].

3.8 Interim analysis

An interim analysis is planned and was specified in Bayesian terms for the ethics submission. It is not described in this paper. [OPEN: decide: brief pointer here plus a companion paper, or a full section added before submission]

4 Reporting

Results will be reported following CONSORT 2025 with the extension for cluster randomised trials [3, 44], including a flow diagram that tracks both practices and participants (Figure 1). Two items of the cluster extension have shaped this plan directly: items 7a and 17a require the intra-cluster correlation to be reported, with an indication of its uncertainty, at the design stage and for each primary outcome; item 13a requires the flow diagram to give the number of clusters as well as the number of individuals at every stage [3].

The guideline for the content of statistical analysis plans that this document follows [4] contains no items specific to clustering; an extension for cluster randomised trials is under development but not yet published [45]. We have therefore added the cluster-specific items ourselves: the estimand and its weighting, the small-sample correction, the intra-cluster correlation, and the two-level description of the sample.

Declarations

Funding. [OPEN: funder and grant number]

Conflicts of interest. [OPEN: to be completed by all authors]

Ethics. [OPEN: committee, approval number, date]

350 **Data and code availability.** The code that reproduces the sample size and power calculations
351 reported here accompanies this paper. Analysis code for the trial will be published together
352 with the trial results. [\[OPEN: repository URL\]](#)

353 **Author contributions.** [\[OPEN: CRediT statement\]](#)

Table 1: Sample size per arm required to detect a 2-point difference in RMDQ at 18 weeks with 80% power and two-sided $\alpha = 0.05$, by intra-cluster correlation. Based on a pooled SD of 5.2 points, adjustment for the baseline score (assumed correlation 0.6), 14 analysed participants per practice and a coefficient of variation of practice size of 0.65.

	Intra-cluster correlation			
	0	0.01	0.03	0.05
Design effect (eq. 1)	1.00	1.19	1.56	1.93
Analysed participants per arm	69	82	107	133
Recruited per arm (10% attrition)	76	91	119	148

Table 2: Power to detect a 2-point difference in RMDQ at 18 weeks, two-sided $\alpha = 0.05$, with a t reference distribution on $k_1 + k_2 - 2$ degrees of freedom. Upper panel: the existing 14 practices with 200 analysed participants in total. Lower panel: balanced designs with 14 analysed participants per practice, so that the total sample grows with the number of practices.

		Intra-cluster correlation		
		0.01	0.03	0.05
Practices (int./ctrl.)	Analysed N			
<i>14 practices, N fixed at 200</i>				
5 / 9	200	0.77	0.65	0.56
6 / 8	200	0.80	0.68	0.59
7 / 7	200	0.80	0.69	0.60
<i>Balanced, 14 participants per practice</i>				
5 / 5	140	0.62	0.51	0.43
6 / 6	168	0.73	0.61	0.52
7 / 7	196	0.80	0.69	0.60
8 / 8	224	0.86	0.76	0.66
9 / 9	252	0.90	0.81	0.72
10 / 10	280	0.93	0.86	0.77

Table 3: Baseline characteristics (shell). Panel A describes the clusters, panel B the participants.

Characteristic	Intervention	Control	All
<i>Panel A — practices</i>			
Number of practices	xx	xx	xx
Therapists per practice, median (IQR)	xx (xx to xx)	xx (xx to xx)	xx (xx to xx)
Participants per practice, median (IQR)	xx (xx to xx)	xx (xx to xx)	xx (xx to xx)
Urban setting, n (%)	xx (xx)	xx (xx)	xx (xx)
<i>Panel B — participants</i>			
Number of participants	xx	xx	xx
Age, mean (SD)	xx (xx)	xx (xx)	xx (xx)
Female, n (%)	xx (xx)	xx (xx)	xx (xx)
Duration of current episode	xx (xx)	xx (xx)	xx (xx)
RMDQ, mean (SD)	xx (xx)	xx (xx)	xx (xx)
Pain intensity, mean (SD)	xx (xx)	xx (xx)	xx (xx)
[OPEN: complete]			

Table 4: Analysis of the primary outcome (shell). All three analyses estimate the participant-average difference in RMDQ at 18 weeks.

Analysis	Intervention mean (SD)	Control mean (SD)	Difference (95% CI)	<i>p</i>
Mixed model, Kenward–Roger (primary)	xx (xx)	xx (xx)	xx (xx to xx)	.xx
Cluster-level, size-weighted	—	—	xx (xx to xx)	.xx
Exact randomisation test	—	—	xx (xx to xx)	.xx
Cluster-average effect (unweighted)	—	—	xx (xx to xx)	.xx
Intra-cluster correlation (95% CI)	xx (xx to xx)			

Table 5: Analysis of secondary outcomes (shell).

Outcome	Intervention	Control	Effect (95% CI)	<i>p</i>
[OPEN: list]	xx (xx)	xx (xx)	xx (xx to xx)	.xx

CONSORT flow diagram for cluster randomised trials (shell): practices screened, randomised and analysed in the upper tier; participants screened, recruited, followed up and analysed in the lower tier, within practices. [OPEN: draw]

Figure 1: Flow of practices and participants through the trial.

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